

Ch 05: Page Faults

xv6 a simple Unix-like teaching Operating System

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1 Lazy Allocation



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Page Fault Exception

- RISC-V CPU raises a page fault exception when
 - virtual address (VA) is used that has no mapping in the page table
 - VA has a mapping whose PTE_V flag is clear
 - VA permission (PTE_R, PTE_W, PTE_X, PTE_U) forbid the operation
 - e.g. a store on a page not marked PTE_W
 - e.g. any operation in user mode on a page that is not marked PTE_U

The combination of page tables and page faults is a powerful tool for an OS

- Give kernel level of indirection between virtual and physical addresses
- Kernel can control structure and content of address spaces
- Allow kernel to intercept loads and stores



Kernel can use these capabilities to increase efficiency

- copy-on-write fork allows kernel to transparently share memory between parent and child
- memory-mapped files, where kernel uses paging to cause a file's contents to appear in an application's address space
- lazy memory allocation, only allocate and load pages a program actually reads and writes
 - xv6 implements lazy allocation, but not the other two



Lazy Allocation

xv6's **lazy allocation** has 2 parts

- 1 When an application calls `sbrk` with the flag `SBRK_LAZY` the kernel simply notes the increase in size of the process address space, but it does not allocate physical memory and does not create PTEs yet for the new range of virtual addresses.
- 2 On a page fault on one of these new addresses, kernel allocates a page of physical memory and maps it into the page table.

Advantages:

- Convenient for applications, they can request worst case usage but not pay the cost if don't need all that memory
- Without lazy allocation, all work to grow a lot of memory happens up front, can be notably bad performance hit
 - On the other hand, page faults incur extra overhead that, if the memory is needed, will be much more work than just doing the work up front.



Summary

- RISC-V raises a page fault exception when VA is not yet mapped in page table, or when PTE_V flag is clear, or when permissions don't allow the operation asked for
- Page tables and page table exceptions are tools OS uses to implement many features
 - virtual memory
 - lazy memory allocation
 - copy on write process forks
- xv6 implements lazy memory allocation
 - when application requests dynamic memory (using `sbrk`) can request `SBRK_LAZY` and kernel simply notes increase in virtual address space size
 - on reference to unmapped VA, xv6 allocates a page of physical memory and maps it into the page table



References



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